Experiment No -10(b)
Verification of Time Invariance of a Discrete System

AIM: Verify the Time Invariance of a given Discrete System.

Software Required: Mat lab software
Theory:

TIME INVARIENT SYSTEMS(TI):

A system is called time invariant if its input – output characteristics do not change with time

X(t)---- input : Y(t) ---output
X(t-k) -----delay input by k seconds : Y(t-k) --------Delayed output by k seconds

If Y(t)=T[X(t)] then Y(t-k)=T[X(t-k)] then system is time invariant system.
Program:

```
% Verification of Time Invariance of a Discrete System
% a)y=x^2(n)          b) y(n)=nx(n)
clc;
clear all;
close all;
n=1:9;
x(n)=[1 2 3 4 5 6 7 8 9];
d=3;
% time delay
xd=[zeros(1,d),x(n)];
%x(n-k)
y(n)=x(n).^2;
yd=[zeros(1,d),y];
%y(n-k)
disp('transformation of delay signal yd:');
disp(yd)dy=xd.^2;
% T[x(n-k)]
disp('delay of transformation signal dy:');
disp(dy)
if dy==yd
disp('given system [y(n)=x(n).^2 ]is time invariant');
else
disp('given system is [y(n)=x(n).^2 ]not time invariant');
end
y=n.*x;
```

```
yd=[zeros(1,d),y(n)];
disp('transformation of delay signal yd:');
disp(yd);
n1=1:length(xd);
dy=n1.*xd;
disp('delay of transformation signal dy:');
disp(dy);
if yd==dy
disp('given system [y(n)=nx(n)]is a time invariant');
else
disp('given system [y(n)=nx(n)]not a time invariant');
end
```

Result: The Time Invariance of a given Discrete System is verified.

Output:

```
transformation of delay signal yd:
     0     0     0     1     4     9    16    25    36    49    64    81

delay of transformation signal dy:
     0     0     0     1     4     9    16    25    36    49    64    81

given system [y(n)=x(n).^2 ]is time invariant
transformation of delay signal yd:
     0     0     0     1     4     9    16    25    36    49    64    81

delay of transformation signal dy:
     0     0     0     4    10    18    28    40    54    70    88   108

given system [y(n)=nx(n)]not a time invariant
```

VIVA QUESTIONS:-

1. Define time invariant system with example?
2. Define time variant system with example?
3. Define LTI system?
4. Give mathematical expression for time invariant system?
5. Give another name for time invariant system and time variant system ?